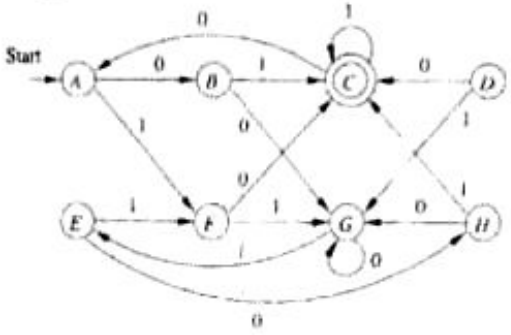
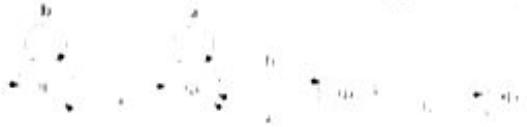
Q. No	CO	Questions	Marks
Q1 a)	1	State the mathemat[illegible]tion of a Finite [illegible]omaton. Also differentiate between NFA and DFA using suitable examples	6
Q1 b)	1	Minimize the following DFA ***with matrix method only*** and verify your answer 	6
Q1 c)	1	What shall be the minimum number of states in a DFA of a language that generates a set of [illegible]ary strings whose last two symbols are always the same? **Support your answer with suitable reasons.**	6
Q2 a)	2	How many strings of length less than 5 will be there in a language generated by the regular expression $(a + b)^* b (x + xy)^*$ **Support your answer with suitable reasons.**	6
Q2 b)	2	Which of the followings law(s) is/are valid for regular expressions? i) L_1: $(L^*)^* = L^*$ ii) L_2: $\Phi^* = \Phi$ iii) L_3: $\varepsilon^* = \varepsilon$ iv) L_4: $L^+ = L^* + \varepsilon$ v) L_5: $(L^*)^* = L^*$ **Support your answer with suitable reasons.**	6
Q2 c)	2	Construct the RE for the given Automaton and verify your answer 	6

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Q3 a)	2	Explain the Chom... ...sification of La... ...details, using suitable examples	6
Q3 b)	3	Construct a CFG for a language described as follows $L_1 = \{ w^a x^b y^c z^d \mid a + b = c + d \}$ **Support your answ... ...with suitable reasons.**	6
Q3 c)	3	Convert following ... FG to CNF $X \rightarrow aYXb \mid YXy$ $Y \rightarrow KZ \mid xay \mid \varepsilon$ $K \rightarrow Ka \mid XY \mid y$ $Z \rightarrow zZ \mid \varepsilon$ **Mention all the step... and the corresponding rules used at appropriate positions.**	6
Q4 a)	4	What is a Push D... Automaton? Explain the concept of configuration of a PDA. Also describe how a PDA can be represented	6
Q4 b)	4	Construct a PDA for the followings i) $L1 = \{ w \in \{0, 1, 2\}^* \mid n_0(w) = ... \}$ ii) $L1 = \{ w \in \{x, y\}^* \mid n_x(w) <= ... \}$ **Validate your answers**	6
Q4 c)	4	Mention the steps to convert the PDA from a CFG along with the advantages as well as disadvantages of this process. Also convert the following CFG to PDA $X \rightarrow aYXb \mid YXy$ $Y \rightarrow KZ \mid xay \mid \varepsilon$ $K \rightarrow Ka \mid XY \mid y$ $Z \rightarrow zZ \mid \varepsilon$	6
Q5 a)	5	Design a TM for the operation f(a, b) = a + b **Verify your answer with suitable examples.**	6
Q5 b)	5	What are primitive recursive functions? Show that $f(a, b) = a^2 + 3b$ is primitive recursive	6
Q5 c)	5	Explain following terms in details with suitable examples i) Recursive Languages vs Recursively Enumerable Languages ii) NP Hard Problems	6

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